THE DERIVED TENSOR PRODUCT WITH THE QUOTIENTS BY THE POWERS OF AN IDEAL AGREES WITH THE PLAIN QUOTIENTS AS A PRO-SYSTEM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 29014-29023 (canonical tag 0G9M) and the counted proof is the proof environment at source lines 29025-29057, which immediately follows it and is its own proof.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which carries the shared Stacks preamble together with the appended definitions of the reference, slogan and history environments; the counted statement and proof do not themselves use those environments, but the span is taken whole so that the file version used for this card is unambiguous.

Editorial note. The proof refers three times, twice to results of this chapter and once to a criterion of another chapter; all three are navigation stubs with their roles recorded in the reasons.

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Allowed standard prerequisites (editorial).

- The inverse system of quotients by the powers of an ideal is pro-isomorphic to the inverse system of Koszul complexes on the powers of a generating set. invoked at proof lines 29026-29031. Replaces the quotient system by the Koszul system, which is a complex of finite free modules sitting in finitely many degrees.
- The Koszul complex on the n-th powers of a set of r generators sits in degrees from -r to 0, so source and target of the comparison have cohomology concentrated in a fixed bounded range. used at proof lines 29032-29037. Makes the pro-isomorphism criterion applicable by bounding the degrees in which the systems live.
- An isomorphism of pro-objects of a derived category is detected on cohomology in each degree, and one may replace a bounded complex by a bounded above complex of finite free modules. used at proof lines 29037-29045. Reduces the statement to a pro-isomorphism of modules in each degree for a convenient representative of the complex.

Source reference. Original lemma 0921: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27172–27172. The reference is invoked at proof lines 29029-29031 to replace the system of quotients by the Koszul system; every later step of the proof uses the replacement and stays inside the counted span.

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Source reference. Original lemma 0G3D: sources/stacks-chow/tags.txt, source lines 20088–20088. The reference is the criterion that lets a pro-isomorphism of complexes be checked on cohomology in each degree, applied at proof lines 29037-29038; the degree bound that makes it applicable is established inside the counted span.

Source reference. Original lemma 0EGU: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28867–28867. The reference is applied at proof lines 29051-29056 to identify both sides of the comparison in each degree with the pro-system of quotients of the cohomology; the reductions that bring the statement into that form are inside the counted span.

Lemma 0.2. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let $M^\bullet$ be a bounded complex of finite A-modules. The inverse system of maps*

$$M^\bullet \otimes^{\mathbf{L}}_A A/I^n \longrightarrow M^\bullet/I^n M^\bullet$$

defines an isomorphism of pro-objects of $D(A)$.

Proof. Say $I = (f_1, \ldots, f_r)$. Let $K_n \in D(A)$ be the object represented by the Koszul complex on $f_1^n, \ldots, f_r^n$. Recall that we have maps $K_n \to A/I^n$ which induce a pro-isomorphism of inverse systems, see Lemma 0921. Hence it suffices to show that

$$M^\bullet \otimes^{\mathbf{L}}_A K_n \longrightarrow M^\bullet/I^n M^\bullet$$

defines an isomorphism of pro-objects of $D(A)$. Since K_n is represented by a complex of finite free A-modules sitting in degrees $-r, \ldots, 0$ there exist $a, b \in \mathbf{Z}$ such that the source and target of the displayed arrow have vanishing cohomology in degrees outside $[a, b]$ for all n. Thus we may apply Derived Categories, Lemma 0G3D and we find that it suffices to show that the maps

$$H^i(M^\bullet \otimes^{\mathbf{L}}_A A/I^n) \to H^i(M^\bullet/I^n M^\bullet)$$

define isomorphisms of pro-systems of A-modules for any $i \in \mathbf{Z}$. To see this choose a quasi-isomorphism $P^\bullet \to M^\bullet$ where $P^\bullet$ is a bounded above complex of finite free A-modules. The arrows above are given by the maps

$$H^i(P^\bullet/I^n P^\bullet) \to H^i(M^\bullet/I^n M^\bullet)$$

These define an isomorphism of pro-systems by Lemma 0EGU. Namely, the lemma shows both are isomorphic to the pro-system $H^i/I^n H^i$ with $H^i = H^i(M^\bullet) = H^i(P^\bullet)$. □